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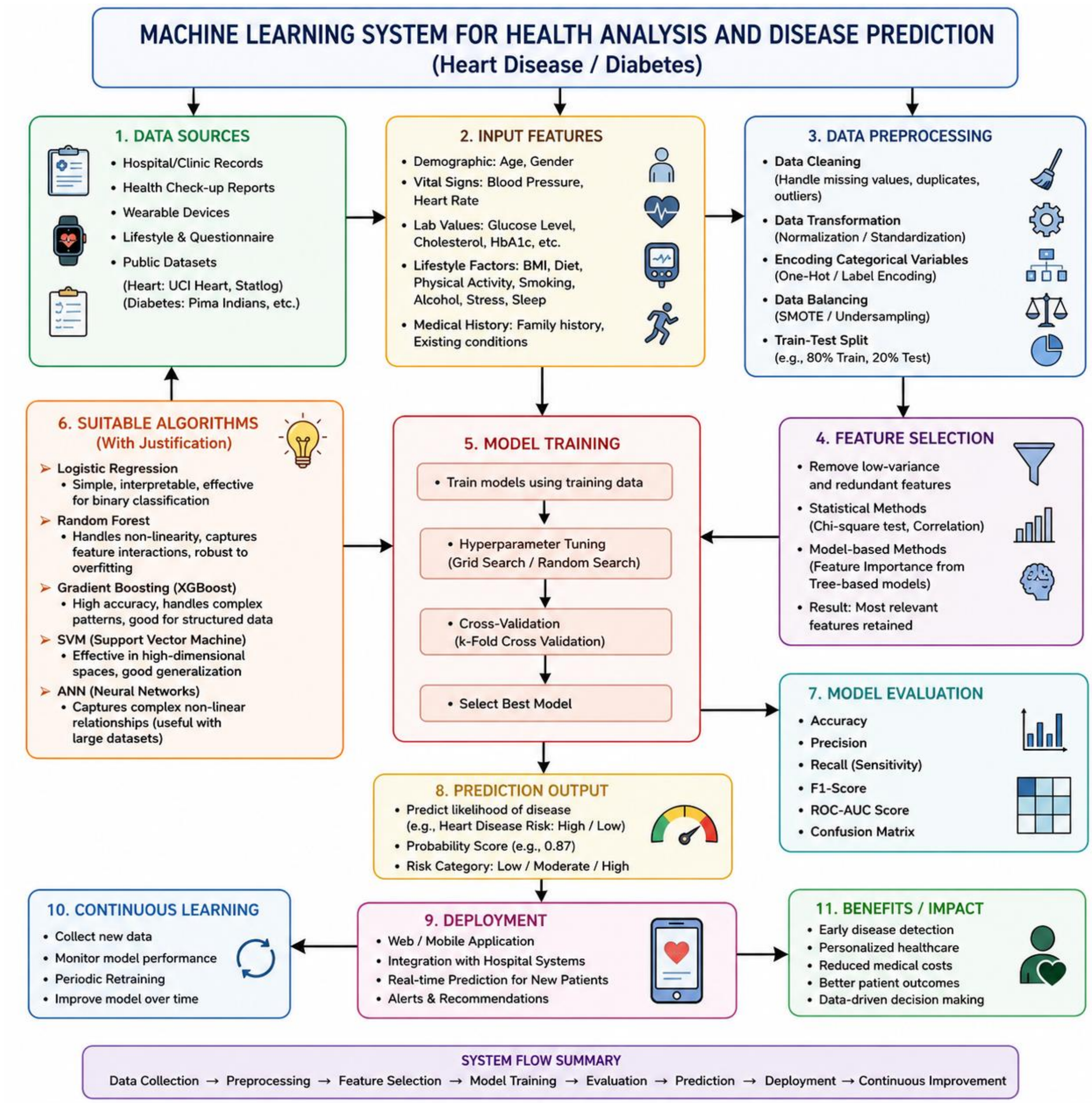
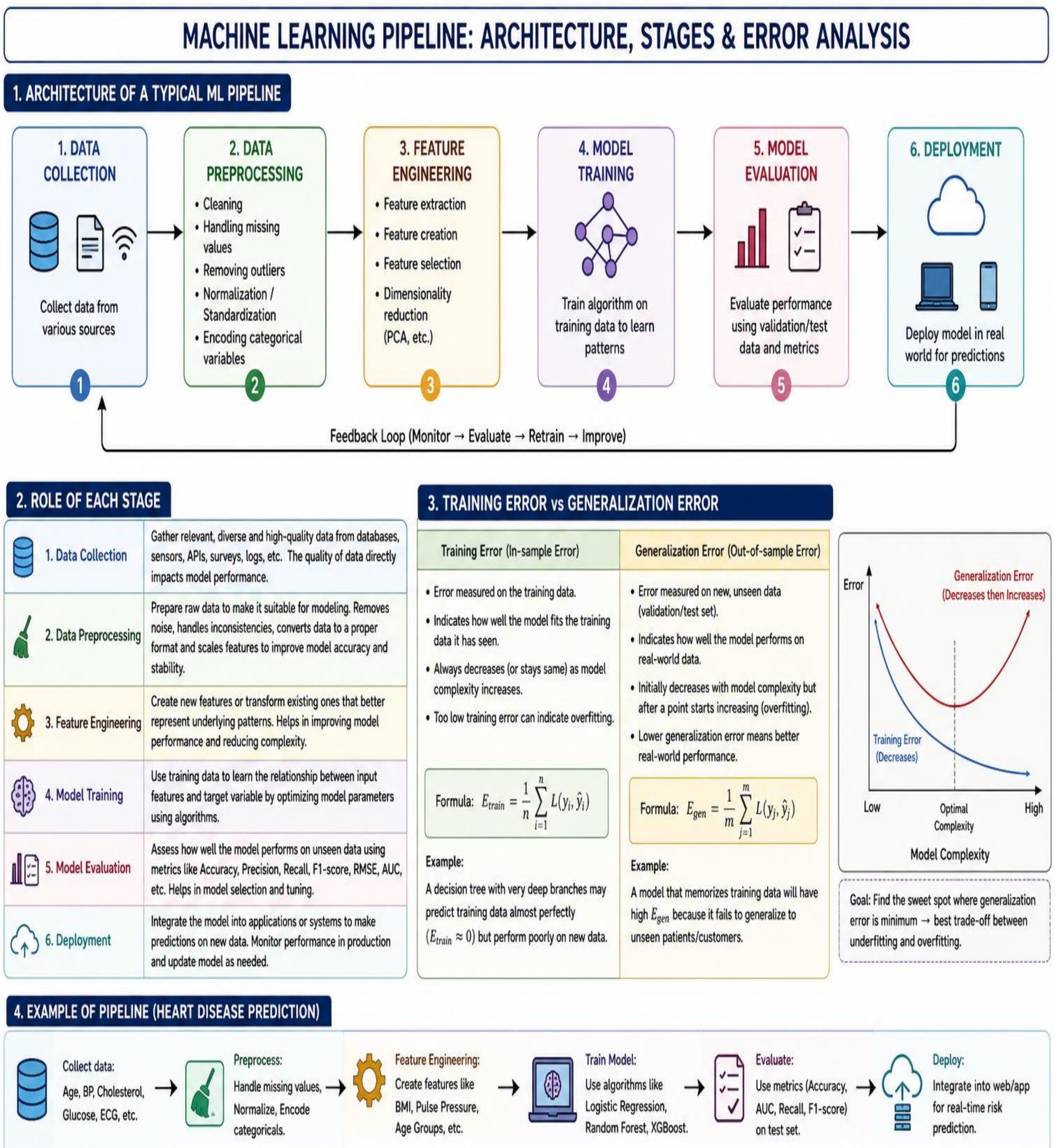
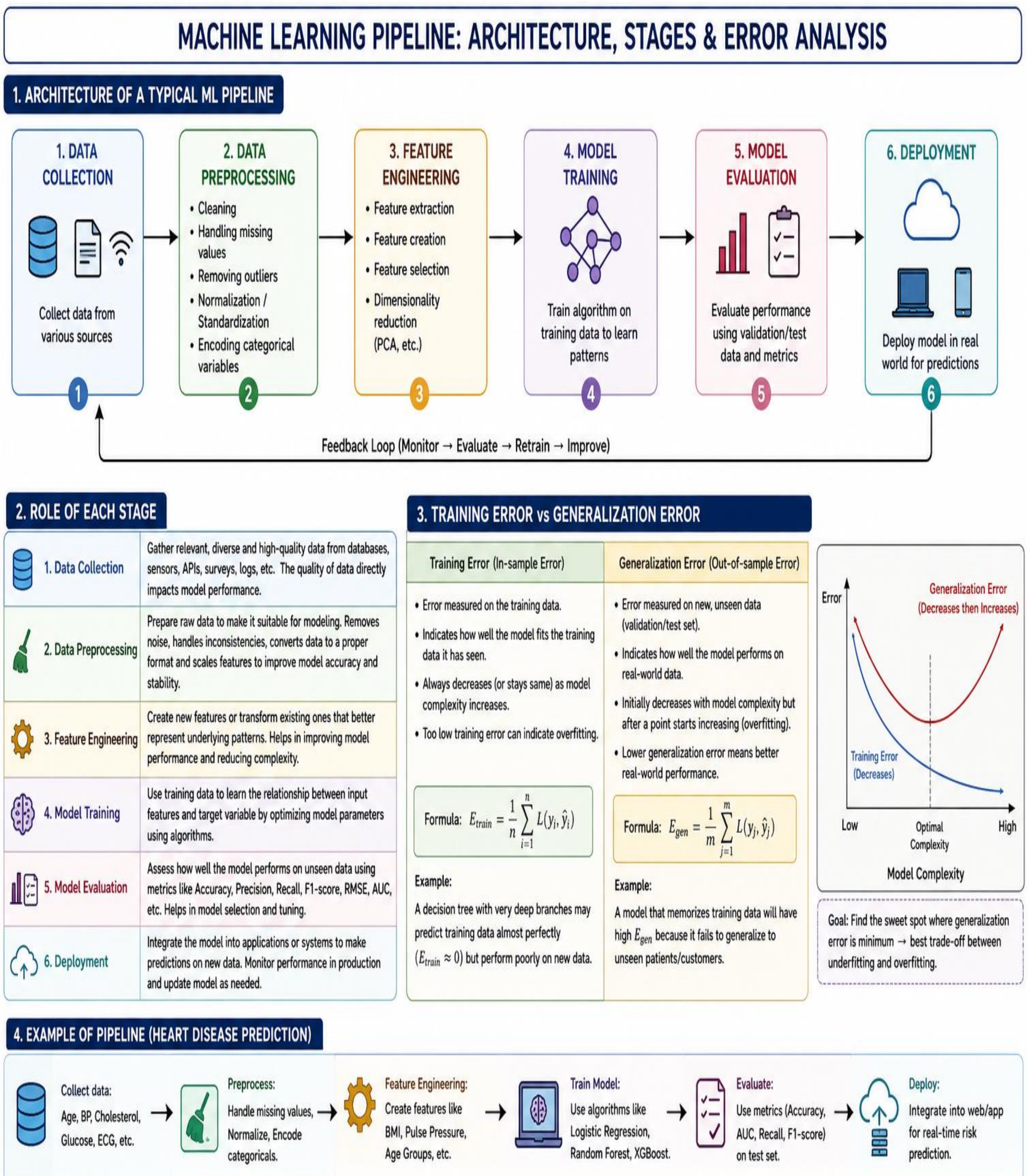
10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. I want concept map



11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.



12. Design a machine learning model for spam email classification using labeled data. (5 Marks)(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.



13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport.(a) Explain the initialization of S (specific boundary) and G (general boundary).(b) Show how S and G change with each training example.(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

13. Candidate–Elimination Algorithm

Dataset (Attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast → Target Concept: EnjoySport)

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
x_1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
x_2	Sunny	Warm	High	Strong	Warm	Same	Yes
x_3	Rainy	Cold	High	Strong	Warm	Change	No
x_4	Sunny	Warm	High	Strong	Cool	Change	Yes
x_5	Sunny	Cold	Normal	Strong	Cool	Change	No
x_6	Rainy	Warm	Normal	Strong	Warm	Same	Yes

(a) INITIALIZATION OF S AND G

1. **Specific boundary (S):**
   Most specific hypothesis (contains only one hypothesis)
   All attributes are set to the most specific value Φ (null symbol).

   $$S_0 = \langle \Phi, \Phi, \Phi, \Phi, \Phi, \Phi \rangle$$

2. **General boundary (G):**
   Most general hypothesis (set of hypotheses)
   Each attribute can take any value from its domain.

   $$G_0 = \{ \langle ?, ?, ?, ?, ?, ? \rangle \}$$

   where ? represents "any value".

Legend

Φ : Most specific value (no value)

? : Any value

This ensures that

$$S_0 \subseteq C \subseteq G_0$$

where C is the set of all candidate hypotheses consistent with training examples seen so far.

(b) CHANGE OF S AND G WITH EACH TRAINING EXAMPLE

Step	Training Example (x_i)	Class (EnjoySport)	Specific Boundary (S)	General Boundary (G)
0 (Init.)	–	–	⟨Φ, Φ, Φ, Φ, Φ, Φ⟩	{⟨?, ?, ?, ?, ?, ?⟩}
1	x_1 = ⟨Sunny, Warm, Normal, Strong, Warm, Same⟩	Yes	⟨Sunny, Warm, Normal, Strong, Warm, Same⟩	{⟨?, ?, ?, ?, ?, ?⟩}
2	x_2 = ⟨Sunny, Warm, *High*, Strong, Warm, Same⟩	Yes	⟨Sunny, Warm, ?, Strong, Warm, Same⟩	{⟨?, ?, ?, ?, ?, ?⟩}
3	x_3 = ⟨*Rainy*, *Cold*, *High*, Strong, Warm, *Change*⟩	No	⟨Sunny, Warm, ?, Strong, Warm, Same⟩	{⟨Sunny, ?, ?, ?, ?, ?⟩, ⟨?, Warm, ?, ?, ?, ?⟩, ⟨?, ?, ?, ?, ?, *Same*⟩}
4	x_4 = ⟨Sunny, Warm, *High*, Strong, *Cool*, Change⟩	Yes	⟨Sunny, Warm, ?, Strong, ?, ?⟩	{⟨Sunny, ?, ?, ?, ?, ?⟩, ⟨?, Warm, ?, ?, ?, ?⟩}
5	x_5 = ⟨Sunny, *Cold*, Normal, Strong, *Cool*, Change⟩	No	⟨Sunny, Warm, ?, Strong, ?, ?⟩	{⟨Sunny, Warm, ?, ?, ?, ?⟩}
6	x_6 = ⟨*Rainy*, Warm, Normal, Strong, Warm, *Same*⟩	Yes	⟨?, Warm, ?, Strong, ?, ?⟩	{⟨Sunny, Warm, ?, ?, ?, ?⟩}

Final Version Space:

$$S_{final} = \langle ?, Warm, ?, Strong, ?, ? \rangle$$

$$G_{final} = \{ \langle Sunny, Warm, ?, ?, ?, ? \rangle \}$$

All hypotheses between S_{final} and G_{final} (inclusive) are consistent with all training examples.

(c) COMPARISON WITH FIND-S ALGORITHM

Aspect	Candidate–Elimination	Find–S
Approach	Maintains both a specific boundary (S) and a general boundary (G) to represent the version space.	Maintains only the most specific hypothesis that is consistent with all positive examples.
Hypotheses Haintained	Set of all hypotheses between S and G ($S \subseteq C \subseteq G$).	Single hypothesis h (most specific).
Use of Negative Examples	Uses both positive and negative examples to generalize S and specialize G.	Uses only positive examples; ignores negative examples.
Generalization Capability	More general. Considers boundaries of the concept, so can converge to a more accurate version space.	Less general. May produce overly specific hypothesis (overfitting).
Convergence Speed	Usually slower (maintains sets and updates both boundaries).	Faster (updates a single hypothesis).
Output	Version space (set of possible hypotheses).	Single hypothesis.
Risk of Overfitting	Lower (because negative examples restrict G).	Higher (cannot exclude unseen negative cases).

Key Takeaway:
Candidate–Elimination is more powerful than *Find–S* because it uses both positive and negative examples and returns the entire hypothesis space, leading to better generalization.

Notation:

Φ – most specific value (no value)

? – any value

S – specific boundary

G – general boundary

C – candidate hypothesis space

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)(a) Justify the choice of regression model (Linear Regression, Decision Tree, or RandomForest).(b) Explain feature scaling and handling missing data.(c) Describe how overfitting can be detected and avoided.